Monish Naresh Kumar

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GitHub: github.com/Brokenx25/MONISH-PROJECTS LinkedIn: linkedin.com/in/monish-naresh-kumar

Robotics & AI engineering student (B.Eng., graduating Sept 2026) with a hands-on portfolio spanning autonomous navigation, computer vision, speech systems, and multi-robot coordination — built end to end in Python and C++. Comfortable taking a robotics problem from perception through control to a working demo, individually and in teams.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, SQL

Robotics & AI: Computer vision, speech recognition, autonomous navigation, human–robot interaction, embedded/hardware prototyping

Simulation & Tools: Webots, CoppeliaSim, Git, HTML/CSS

PROJECTS

- **Medical Robot Dog Assistant (Final-Year Project) —** Building an autonomous quadruped that continuously monitors patients and navigates a care environment, aimed as a lower-cost alternative to trained medical service dogs. *Stack: Python, computer vision, autonomous navigation*
- **Synchronized Multi-Robot Performance —** Designed a controller that coordinates multiple Alpha Mini humanoid robots to perform actions in sync, triggered on demand by anyone through a QR-code-accessible web app. *Stack: Python, web app, multi-robot control*
- **Teaching Assistant Robot (Team Project) —** Programmed an Alpha Mini humanoid to act as an interactive teaching assistant that helps students work through learning difficulties; delivered in a small team. *Stack: Python, HRI, speech*
- **Autonomous Path-Following Car —** Built a self-driving vehicle that follows a planned route while reacting in real time to obstacles and hazards, integrating perception with reactive control. *Stack: Python, computer vision, control*
- **Computer Vision & Speech Pipelines —** Developed a vision pipeline detecting faces, hand gestures, and body pose, plus a speech pipeline that identifies both the speaker and the spoken words. *Stack: Python, OpenCV, speech recognition*
- **Autonomous Robotics Simulations (Webots & CoppeliaSim) —** Simulated a robot exhibiting 10 distinct autonomous behaviours in Webots, and modelled a real-world minefield in CoppeliaSim to analyse hazards and plan safe navigation paths. *Stack: Webots, CoppeliaSim*
- **Line-Following Robot (Built from Scratch) —** Designed and assembled a two-wheeled autonomous robot end to end, covering both the hardware build and the control logic. *Stack: Embedded, C/C++*

EDUCATION

Bachelor of Engineering in Robotics and AI Expected Sept 2026
PSB Academy, Singapore

VOLUNTEERING

Event Volunteer, PSB Academy 2026

- Supported delivery of campus events and programs, coordinating logistics with fellow volunteers and gathering participant feedback to improve future events.

LANGUAGES

English (Advanced, C1) • Tamil (Upper-Intermediate, B2) • Hindi (Intermediate, B1)